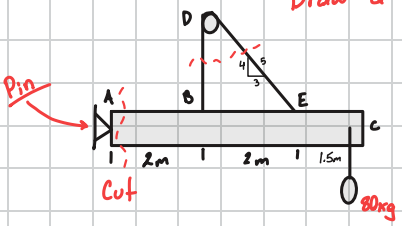
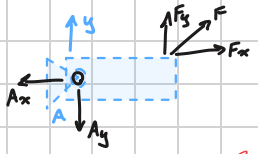
Free body diagram

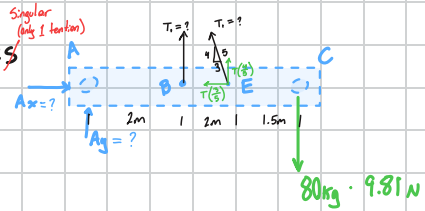
Draw a FBD



① pin



② cables

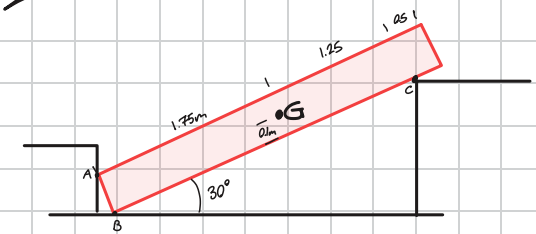


Solving

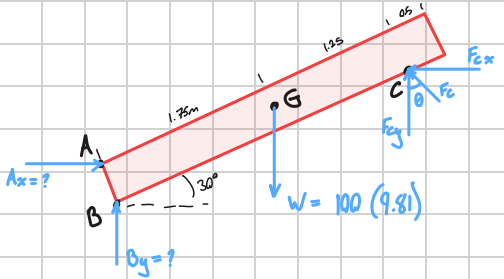
① find first unknown with moment

② solve for other unknowns

Example

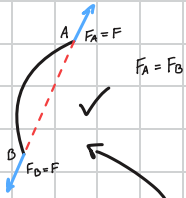


FBD



two force member





two forces with the line being the same with no moment

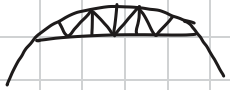
trusses are an assembly of two force members

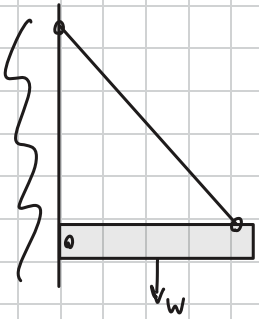
ex

bridges

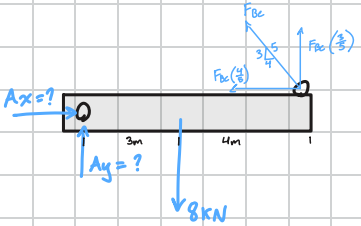


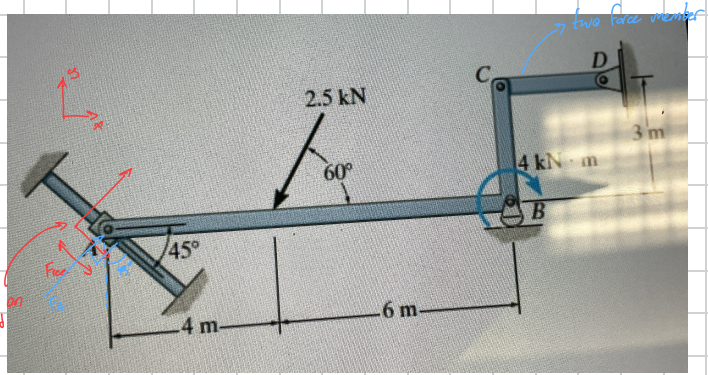
large ceilings



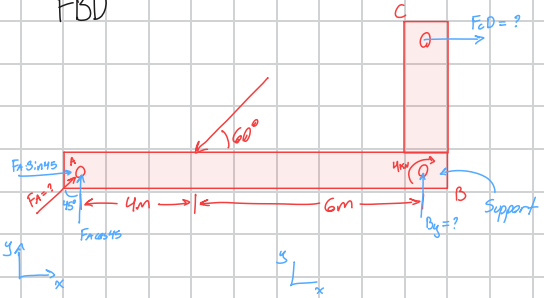


FBD

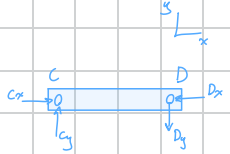




FBD



F_A, B_y, F_{CD} ??



$\rightarrow \sum F_x = 0$ $\quad C_x + D_x = 0$ $\quad C_y + D_y = 0$

$C_x = -D_x$ $\quad C_y = -D_y$

$\circlearrowleft \Sigma M$

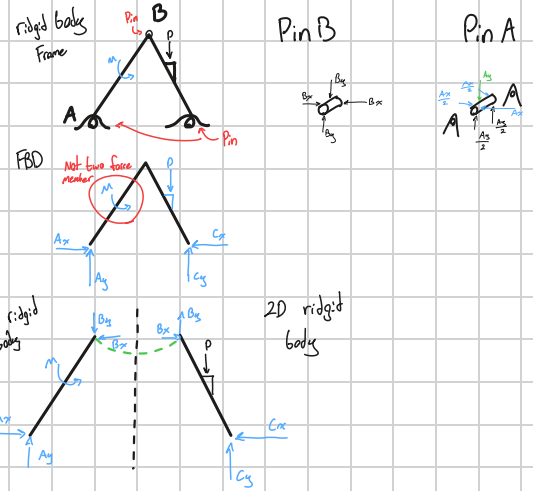
Three force member

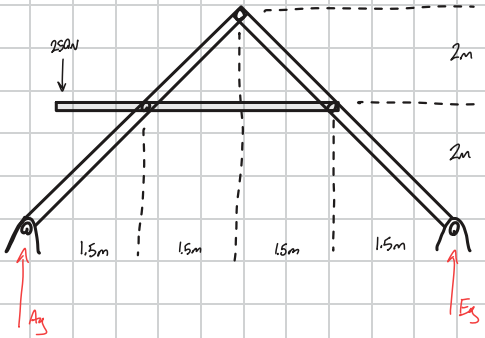


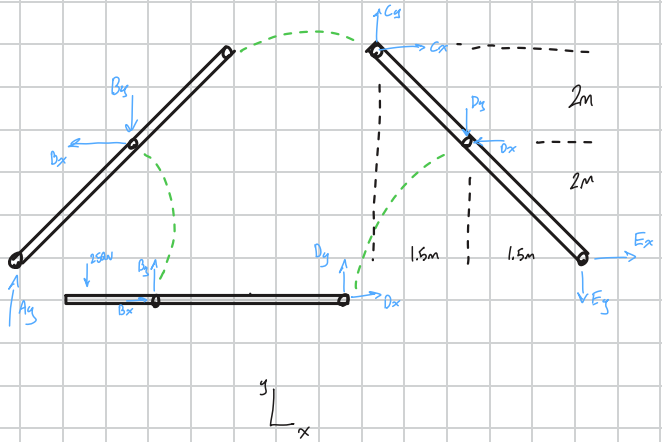
Free body diagram

a) internal forces in members

multi-member bodies




250N
2m
2m
1.5m
1.5m
1.5m
1.5m
A_y
E_y


C_y
C_x
B_y
B_x
D_y
D_x
2m
2m
250N
B_y
B_x
D_y
D_x
1.5m
1.5m
E_x
A_y
E_y

y
x